

EFQAT: AN EFFICIENT FRAMEWORK FOR QUANTIZATION-AWARE TRAINING

Saleh Ashkboos^{1*} Bram Verhoef² Torsten Hoefer¹ Evangelos Eleftheriou² Martino Dazzi²

ABSTRACT

Quantization-aware training (QAT) schemes have been shown to achieve near-full precision accuracy. They accomplish this by training a quantized model for multiple epochs. This is computationally expensive, mainly because of the full precision backward pass. On the other hand, post-training quantization (PTQ) schemes do not involve training and are therefore computationally cheap, but they usually result in a significant accuracy drop. We address these challenges by proposing EfQAT, which generalizes both schemes by optimizing only a subset of the parameters of a quantized model. EfQAT starts by applying a PTQ scheme to a pre-trained model and only updates the most critical network parameters while freezing the rest, accelerating the backward pass. We demonstrate the effectiveness of EfQAT on various CNNs and Transformer-based models using different GPUs. Specifically, we show that EfQAT is significantly more accurate than PTQ with little extra compute. Furthermore, EfQAT can accelerate the QAT backward pass between 1.44-1.64x while retaining most accuracy.

1 INTRODUCTION

Large-scale neural networks are at the core of many advances in Artificial Intelligence. Model scaling is of prime importance and a key enabler in achieving ever-increasing performance on a variety of cognitive tasks. Indeed, state-of-the-art models have moved from 60M parameters in 2012 (Krizhevsky et al., 2017) to 540B in 2022 (Chowdhery et al., 2022), meanwhile showing a strong correlation between the increase in capabilities of the Deep Neural Network (DNNs) models and the number of parameters. Consequently, training these increasingly accurate and general models implies increasing hardware requirements in terms of memory, compute, and energy (OpenAI, 2018; Strubell et al., 2019). Given the current trend, future huge models may be infeasible to train with currently available systems.

Previous work has addressed this resource challenge by introducing methods to reduce the model size. These methods often rely on quantization (Gholami et al., 2021) and sparsification (Hoefer et al., 2021). The former tries to reduce the computational and memory cost of neural network training and inference by allocating a lower number of bits to different network variables (e.g. weights and activations), whereas the latter aims at retaining the most critical parameters and pruning the rest. Quantization has gained more

^{*}Work done during an internship at Axelera AI. [ETH Zurich](#) ²Axelera AI. Correspondence to: Saleh Ashkboos <saleh.ashkboos@inf.ethz.ch>.

Preprint.

Quantization Schemes	Near FP Accuracy	High Performance
Post-Training Quantization (PTQ)	✗	✓
Quantization-Aware Training (QAT)	✓	✗
EfQAT	✓	✓

Table 1. Different quantization schemes: EfQAT achieves higher accuracy than PTQ and is faster than QAT. FP refers to the full precision model.

attention as it is easier to be applied in practice with the available hardware. As the dynamic range of activations and weights varies considerably across the network layers, the main challenge of quantizing neural networks lies in mapping real-valued, continuous operands and parameters to a discrete integer-valued domain without significant distortion.

Post-training quantization (PTQ) schemes quantize models in a greedy fashion. Specifically, these algorithms usually optimize an auxiliary loss function representing the distance between the quantized and unquantized operands and parameters in the network. The optimization is applied on a *per-layer* basis using only a few training samples, known as the calibration set. Although these approaches are computationally inexpensive, they often decrease network accuracy.

Quantization-aware training (QAT) is another class of neural network quantization algorithms that tries to make the above transformation towards lower precision networks easier by training the quantized network directly. These